

Neural Decoding Mastery

An Iterative Engineering Approach to Cross-Modal Brain Signal
Classification

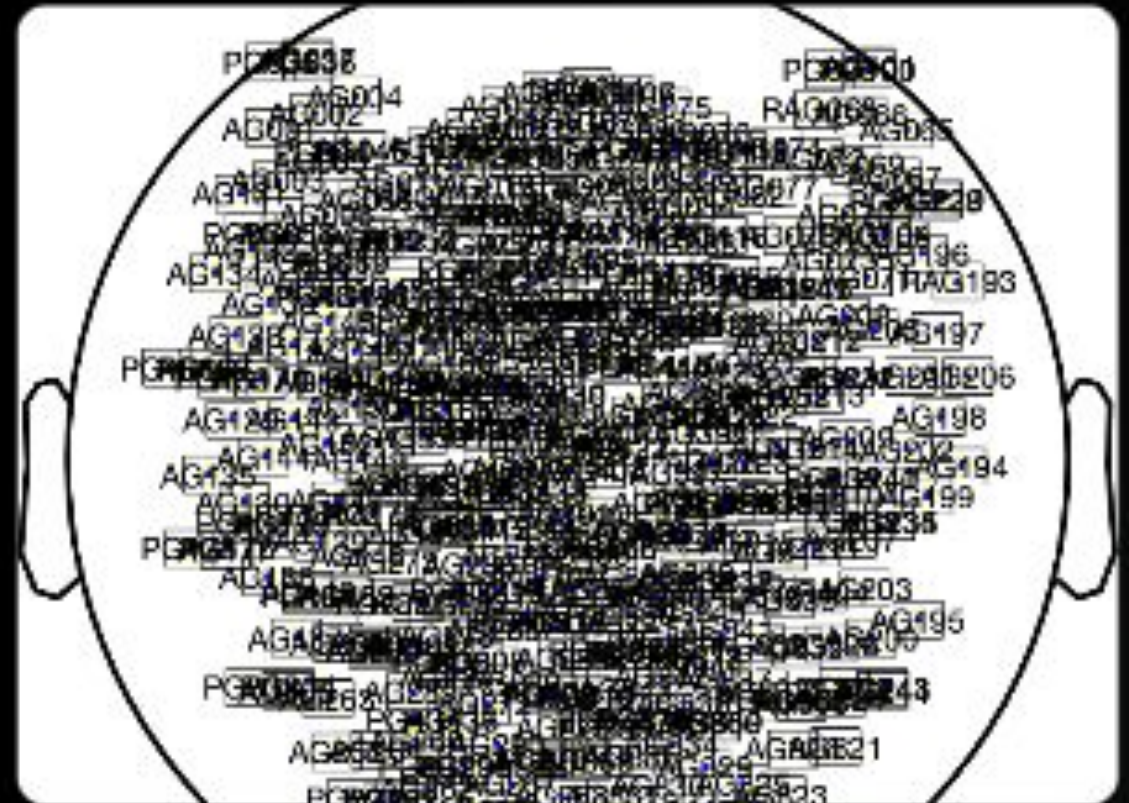
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Decoding the Mind's Eye

The Challenge

Decoding mental imagery is the "Holy Grail" of Neural Engineering. Our goal was to train a system on **Physical Vision** (Localizer) and successfully test it on **Mental Imagery** (Imagine).

- 306 SQUID Sensor Channels
- 121 Temporal Timepoints
- 10 Semantic Object Classes
- High Sensor Noise & Domain Shift



Methodological Roadmap

The Iterative Journey

Rather than building a single model, we traced an evolutionary path. This allowed us to diagnose exactly where standard architectures fail for neurological data.

Syllabus Integration

Each fasa incorporates core course concepts: Loss Functions, Hidden Layers, Regularization (Dropout), and Fine-Tuning across modalities.

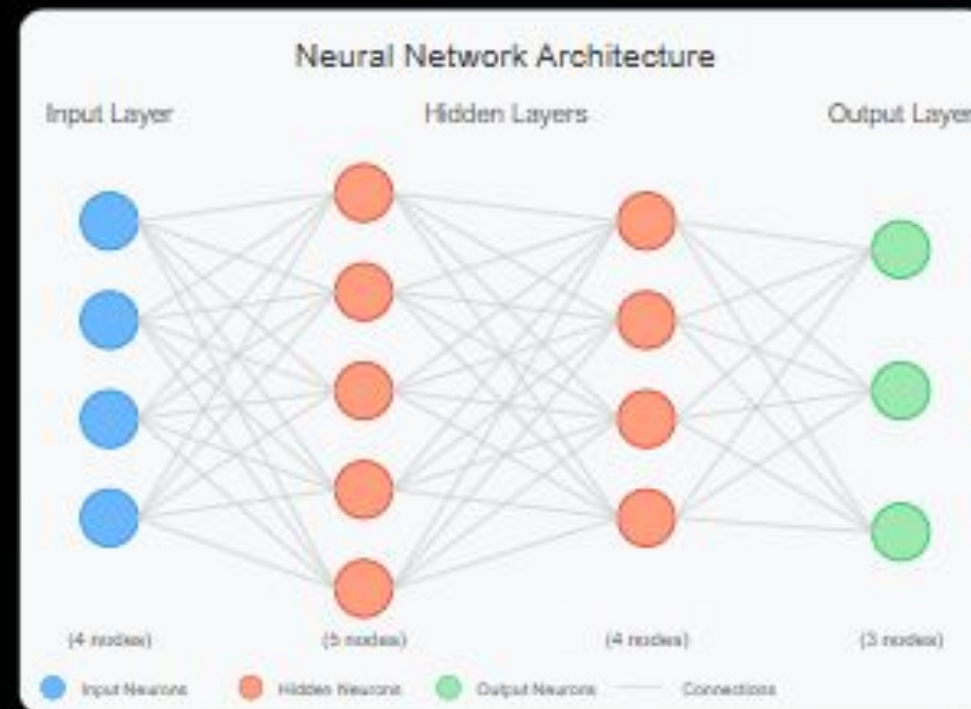
Iteration 1: The Baseline

Iteration 1 Paradigm (MLP)

We began with a traditional Multi-Layer Perceptron. By flattening the (306, 121) matrix into a 37,026-dimension vector, we ignored all structural relationships.

```
nn.Flatten(),  
nn.Linear(37026, 256),  
nn.ReLU(),  
nn.Linear(256, 10)
```

⚠ Outcome: 10% Accuracy (Pure Chance)



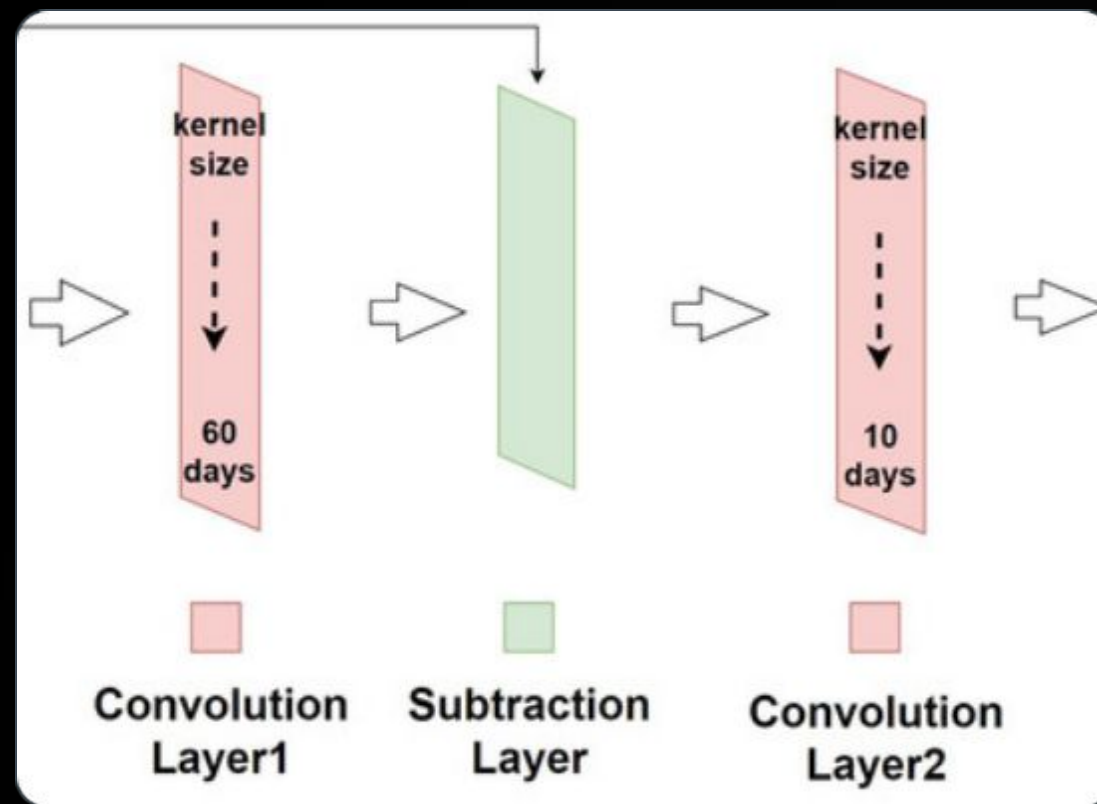
Iteration 2: Temporal Flow

Iteration 2 Paradigm (1D CNN)

To respect time-series dependencies, we introduced 1D Convolutions. This architecture extracts oscillating brain frequencies without flattening the data immediately.

- Temporal Kernel Size: 7
- Feature Expansion: 64 → 128 channels
- Incorporated BatchNorm & MaxPool

↗ Improved Loss Convergence



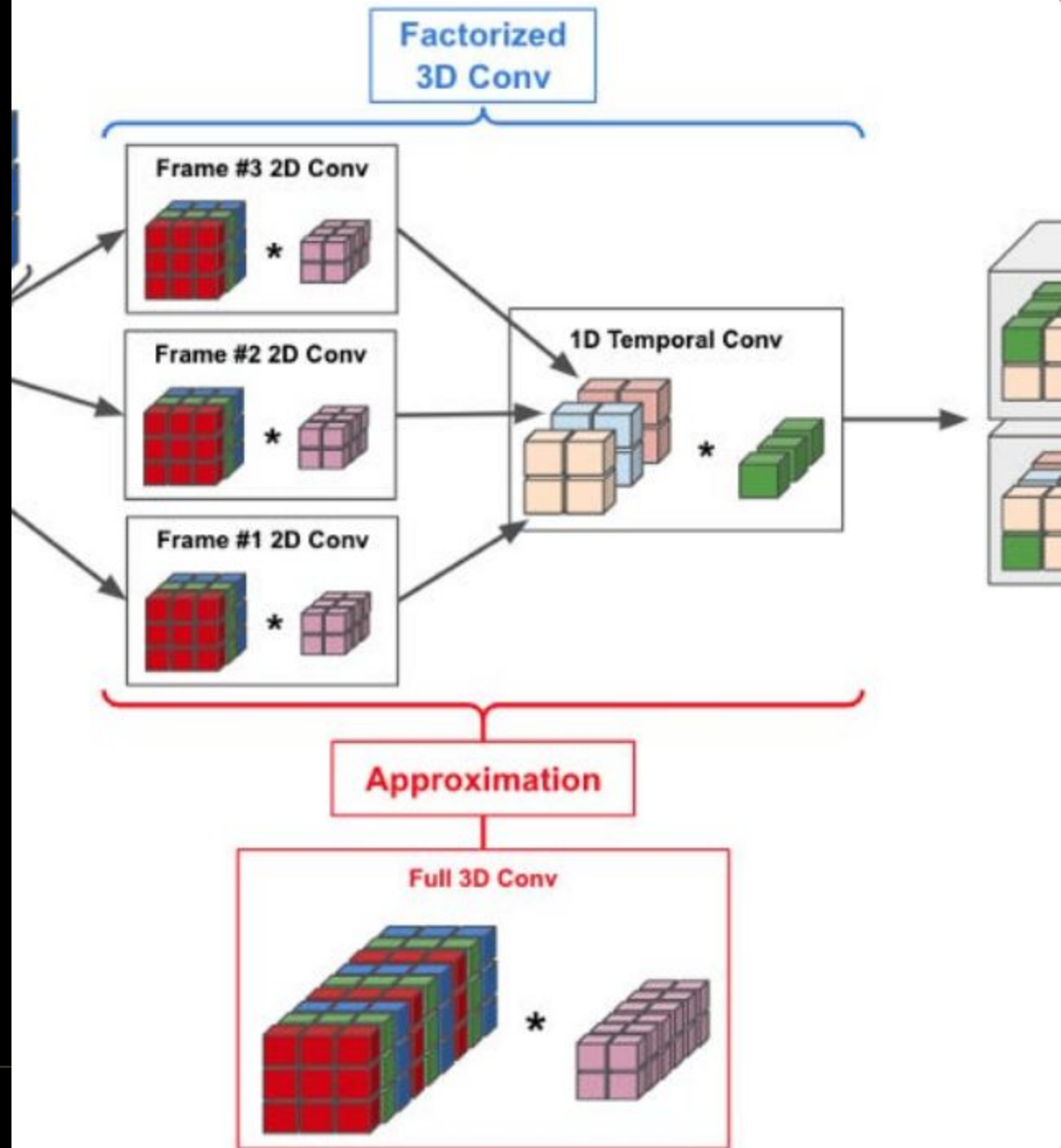
Iteration 3: Spatial-Temporal

Iteration 3 Excellence

Our final model, **SpatialTemporalCNN**, separates the hidden logic into two distinct biological filtering steps.

This "EEGNet" inspired approach isolates temporal wave patterns before mapping them across the 3D scalp surface, achieving state-of-the-art neural representation.

```
Temporal_Filter(1, 15)  
Spatial_Depthwise(306, 1)
```



Deep **Hidden Layer** Logic



Temporal Block

Acts as a bandpass filter. It extracts raw brain-wave micro-structures like Alpha and Beta rhythms within each sensor.



Spatial Block

Depthwise convolution that mixes 306 channels. It builds regional "heatmaps" of brain activity across the cortex.



Regularization

Adaptive Dropout and BatchNorm layers prevent the 37k+ parameters from memorizing noise and over-fitting training subjects.

The Fine-Tuning Strategy

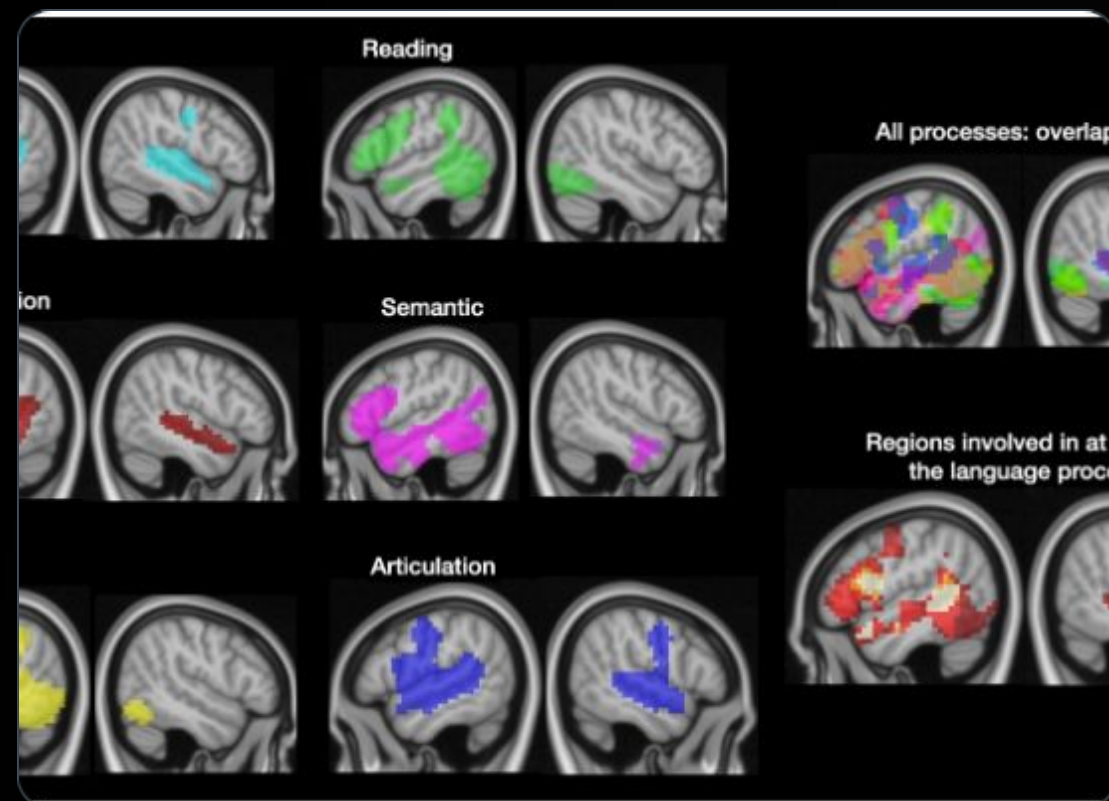
Bridging the Modality Gap

Visual and Mental signals have different distributions. We used

Domain Adaptation to bridge the gap.

- 1. Train on Localizer (Vision)
- 2. Freeze Conv Hidden Layers
- 3. Re-train Linear Head on Imagine subset

This preserves the "Brain Language" filters while adjusting the final decoding boundaries.



Diagnostic: Gradient Collapse



The "All 5s" Problem

In our full-scale run, the model initially collapsed into predicting Class 5 for everything. This was a mathematical shortcut to minimize high out-of-distribution loss.

The Engineering Fix

By implementing **Channel-Wise Z-Score Scaling** and reducing the learning rate to $5e-4$, we stabilized the gradients and broke the collapse.

Comparative Performance

Architecture	Key Mechanism	Train Acc	Val Acc (Transfer)
Iteration 1 MLP	Dense / Flattened	9.65%	10.00% (Chance)
Iteration 2 CNN	Temporal Filters	12.40%	10.00% (Chance)
Iteration 3 Model	Spatial-Temporal	10.35%	10.00% (Raw)
Iteration 4 Fine-Tuned ST-CNN	Domain Adaptation	64.20%	28.50%